



Figure 10.2 Sample output of `bernoullitraces.m`, including the deterministic standard error graphs. The graph shows how at any given step, about two thirds of the sample means are within one standard error of the true mean.

traces, such that M_k is within one standard error of the expected value p . Graph $T_k = M_k/m$ as a function of k . Explain your results.

Further Reading: [Dur94] contains concise, rigorous presentations and proofs of the laws of large numbers. [WS01] covers parameter estimation for both scalar and vector random variables and stochastic processes.

Problems

Difficulty: ● Easy ■ Moderate ◆ Difficult ◆◆ Experts Only

10.1.1 ● $X_1, \dots, X_n$ is an iid sequence of exponential random variables, each with expected value 5.

- What is $\text{Var}[M_9(X)]$, the variance of the sample mean based on nine trials?
- What is $P[X_1 > 7]$, the probability that one outcome exceeds 7?
- Use the central limit theorem to estimate $P[M_9(X) > 7]$, the probability that the sample mean of nine trials exceeds 7.

10.1.2 ● $X_1, \dots, X_n$ are independent uniform random variables with expected value $\mu_X = 7$ and variance $\text{Var}[X] = 3$.

- What is the PDF of X_1 ?
- What is $\text{Var}[M_{16}(X)]$, the variance of the sample mean based on 16 trials?
- What is $P[X_1 > 9]$, the probability that one outcome exceeds 9?
- Would you expect $P[M_{16}(X) > 9]$ to be bigger or smaller than $P[X_1 > 9]$?

To check your intuition, use the central limit theorem to estimate $P[M_{16}(X) > 9]$.

10.1.3 ■ X is a uniform $(0, 1)$ random variable. $Y = X^2$. What is the standard error of the estimate of μ_Y based on 50 independent samples of X ?

10.1.4 ■ Let $X_1, X_2, \dots$ denote a sequence of independent samples of a random variable X with variance $\text{Var}[X]$. We define a new random sequence $Y_1, Y_2, \dots$ as $Y_1 = X_1 - X_2$ and $Y_n = X_{2n-1} - X_{2n}$.

- Find $E[Y_n]$ and $\text{Var}[Y_n]$.
- Find the expected value and variance of $M_n(Y)$.

10.2.1 ● The weight of a randomly chosen Maine black bear has expected value $E[W] = 500$ pounds and standard deviation $\sigma_W = 100$ pounds. Use the Chebyshev inequality to upper bound the probability that the weight of a randomly chosen bear

is more than 200 pounds from the expected value of the weight.

10.2.2 ● For an arbitrary random variable X , use the Chebyshev inequality to show that the probability that X is more than k standard deviations from its expected value $E[X]$ satisfies

$$P[|X - E[X]| \geq k\sigma] \leq \frac{1}{k^2}.$$

For a Gaussian random variable Y , use the $\Phi(\cdot)$ function to calculate the probability that Y is more than k standard deviations from its expected value $E[Y]$. Compare the result to the upper bound based on the Chebyshev inequality.

10.2.3 ■ Elevators arrive randomly at the ground floor of an office building. Because of a large crowd, a person will wait for time W in order to board the third arriving elevator. Let X_1 denote the time (in seconds) until the first elevator arrives and let X_i denote the time between the arrival of elevator $i - 1$ and i . Suppose X_1, X_2, X_3 are independent uniform $(0, 30)$ random variables. Find upper bounds to the probability W exceeds 75 seconds using

- (a) the Markov inequality,
- (b) the Chebyshev inequality,
- (c) the Chernoff bound.

10.2.4 ■ Let X equal the arrival time of the third elevator in Problem 10.2.3. Find the exact value of $P[W \geq 75]$. Compare your answer to the upper bounds derived in Problem 10.2.3.

10.2.5 ■ In a game with two dice, the event *snake eyes* refers to both six-sided dice showing one spot. Let R denote the number of dice rolls needed to observe the third occurrence of *snake eyes*. Find

- (a) the upper bound to $P[R \geq 250]$ based on the Markov inequality,
- (b) the upper bound to $P[R \geq 250]$ based on the Chebyshev inequality,
- (c) the exact value of $P[R \geq 250]$.

10.2.6 ■ Use the Chernoff bound to show that the Gaussian $(0, 1)$ random variable Z satisfies

$$P[Z \geq c] \leq e^{-c^2/2}.$$

For $c = 1, 2, 3, 4, 5$, use Table 4.2 and Table 4.3 to compare the Chernoff bound to the true value: $P[Z \geq c] = Q(c)$.

10.2.7 ■ Use the Chernoff bound to show for a Gaussian (μ, σ) random variable X that

$$P[X \geq c] \leq e^{-(c-\mu)^2/2\sigma^2}.$$

Hint: Apply the result of Problem 10.2.6.

10.2.8 ■ Let K be a Poisson random variable with expected value α . Use the Chernoff bound to find an upper bound to $P[K \geq c]$. For what values of c do we obtain the trivial upper bound $P[K \geq c] \leq 1$?

10.2.9 ■ In a subway station, there are exactly enough customers on the platform to fill three trains. The arrival time of the n th train is $X_1 + \cdots + X_n$ where $X_1, X_2, \dots$ are iid exponential random variables with $E[X_i] = 2$ minutes. Let W equal the time required to serve the waiting customers. Find $P[W > 20]$.

10.2.10 ■ Let $X_1, \dots, X_n$ be independent samples of a random variable X . Use the Chernoff bound to show that $M_n(X) = (X_1 + \cdots + X_n)/n$ satisfies

$$P[M_n(X) \geq c] \leq \left(\min_{s \geq 0} e^{-sc} \phi_X(s) \right)^n.$$

10.3.1 ● Let $X_1, X_2, \dots$ denote an iid sequence of random variables, each with expected value 75 and standard deviation 15.

- (a) How many samples n do we need to guarantee that the sample mean $M_n(X)$ is between 74 and 76 with probability 0.99?
- (b) If each X_i has a Gaussian distribution, how many samples n' would we need to guarantee $M_{n'}(X)$ is between 74 and 76 with probability 0.99?

10.3.2 Let X_A be the indicator random variable for event A with probability $P[A] = 0.8$. Let $\hat{P}_n(A)$ denote the relative frequency of event A in n independent trials.

- Find $E[X_A]$ and $\text{Var}[X_A]$.
- What is $\text{Var}[\hat{P}_n(A)]$?
- Use the Chebyshev inequality to find the confidence coefficient $1 - \alpha$ such that $\hat{P}_{100}(A)$ is within 0.1 of $P[A]$. In other words, find α such that

$$P\left[\left|\hat{P}_{100}(A) - P[A]\right| \leq 0.1\right] \geq 1 - \alpha.$$

- Use the Chebyshev inequality to find out how many samples n are necessary to have $\hat{P}_n(A)$ within 0.1 of $P[A]$ with confidence coefficient 0.95. In other words, find n such that

$$P\left[\left|\hat{P}_n(A) - P[A]\right| \leq 0.1\right] \geq 0.95.$$

10.3.3 $X_1, X_2, \dots$ is a sequence of iid Bernoulli $(1/2)$ random variables. Consider the random sequence $Y_n = X_1 + \dots + X_n$.

- What is $\lim_{n \rightarrow \infty} P[|Y_{2n} - n| \leq \sqrt{n/2}]$?
- What does the weak law of large numbers say about Y_{2n} ?

10.3.4 In communication systems, the error probability $P[E]$ may be difficult to calculate; however it may be easy to derive an upper bound of the form $P[E] \leq \epsilon$. In this case, we may still want to estimate $P[E]$ using the relative frequency $\hat{P}_n(E)$ of E in n trials. In this case, show that

$$P\left[\left|\hat{P}_n(E) - P[E]\right| \geq c\right] \leq \frac{\epsilon}{nc^2}.$$

10.3.5 A factory manufactures chocolate chip cookies on an assembly line. Each cookie is sprinkled with K chips from a very large vat of chips, where K is Poisson with $E[K] = 10$, independent of the number on any other cookie. Imagine you are a chip in the vat and you are sprinkled onto a cookie. Let J denote the number of chips (including you) in your cookie. What is the PMF of J ?

Hint: Suppose n cookies have been made such that N_k cookies have k chips. You are just one of the $\sum_{k=0}^{\infty} kN_k$ chips used in the n cookies.

10.3.6 In this problem, we develop a weak law of large numbers for a correlated sequence $X_1, X_2, \dots$ of identical random variables. In particular, each X_i has expected value $E[X_i] = \mu$, and the random sequence has covariance function

$$C_X[m, k] = \text{Cov}[X_m, X_{m+k}] = \sigma^2 a^{|k|}$$

where a is a constant such that $|a| < 1$. For this correlated random sequence, we can define the sample mean of n samples as

$$M_n = \frac{X_1 + \dots + X_n}{n}.$$

- Use Theorem 9.2 to show that

$$\text{Var}[X_1 + \dots + X_n] \leq n\sigma^2 \left(\frac{1+a}{1-a} \right).$$

- Use the Chebyshev inequality to show that for any $c > 0$,

$$P[|M_n - \mu| \geq c] \leq \frac{\sigma^2(1+a)}{n(1-a)c^2}.$$

- Use part (b) to show that for any $c > 0$,

$$\lim_{n \rightarrow \infty} P[|M_n - \mu| \geq c] = 0.$$

10.3.7 In the Gaussian Movie DataBase (GMDB), reviewers like you rate movies with Gaussian scores. In particular, the first person to rate a movie assigns a Gaussian $(q, 1)$ review score X_1 , where r_0 is the true “quality” of the movie. After n reviews, a movie’s rating is $R_n = \sum_{i=1}^n X_i/n$. Strangely enough, in the GMDB, reviewers are influenced by prior reviews; if after $n-1$ reviews a movie is rated $R_{n-1} = r$, the n th review n will rate the movie X_n , a Gaussian $(r, 1)$ random variable, *conditionally* independent of $X_1, \dots, X_{n-1}$ given $R_{n-1} = r$.

- Find $E[R_n]$.

- (b) Find the PDF $f_{R_n}(r)$. Hint: You may have unresolved parameters in this answer.
- (c) Find $\text{Var}[R_n]$. Hint: Find $E[R_n^2 | R_{n-1}]$.
- (d) Interpret your results as $n \rightarrow \infty$? Does the law of large numbers apply here?

10.4.1 ● When X is Gaussian, verify Equation (10.24), which states that the sample mean is within one standard error of the expected value with probability 0.68.

10.4.2 ■ Suppose the sequence of estimates $\hat{R}_n$ is biased but asymptotically unbiased. If $\lim_{n \rightarrow \infty} \text{Var}[\hat{R}_n] = 0$, is the sequence $\hat{R}_n$ consistent?

10.4.3 ■ An experimental trial produces random variables X_1 and X_2 with correlation $r = E[X_1 X_2]$. To estimate r , we perform n independent trials and form the estimate

$$\hat{R}_n = \frac{1}{n} \sum_{i=1}^n X_1(i) X_2(i),$$

where $X_1(i)$ and $X_2(i)$ are samples of X_1 and X_2 on trial i . Show that if $\text{Var}[X_1 X_2]$ is finite, then $\hat{R}_1, \hat{R}_2, \dots$ is an unbiased, consistent sequence of estimates of r .

10.4.4 ■ An experiment produces random vector $\mathbf{X} = [X_1 \ \cdots \ X_k]'$ with expected value $\boldsymbol{\mu}_{\mathbf{X}} = [\mu_1 \ \cdots \ \mu_k]'$. The i th component of $\mathbf{X}$ has variance $\text{Var}[X_i] = \sigma_i^2$. To estimate $\boldsymbol{\mu}_{\mathbf{X}}$, we perform n independent trials such that $\mathbf{X}(i)$ is the sample of $\mathbf{X}$ on trial i , and we form the vector mean

$$\mathbf{M}(n) = \frac{1}{n} \sum_{i=1}^n \mathbf{X}(i).$$

- (a) Show $\mathbf{M}(n)$ is unbiased by showing $E[\mathbf{M}(n)] = \boldsymbol{\mu}_{\mathbf{X}}$.
- (b) Show that the sequence of estimates $\mathbf{M}_n$ is consistent by showing that for any constant $c > 0$,

$$\lim_{n \rightarrow \infty} P \left[\max_{j=1, \dots, k} |M_j(n) - \mu_j| \geq c \right] = 0.$$

Hint: Let $A_i = \{|M_i(n) - \mu_i| \geq c\}$ and apply the union bound (see Problem 1.3.11) to upper bound $P[A_1 \cup A_2 \cup \cdots \cup A_k]$. Then apply the Chebyshev inequality.

10.4.5 ♦ Given the iid samples $X_1, X_2, \dots$ of X , define the sequence $Y_1, Y_2, \dots$ by

$$Y_k = \left(X_{2k-1} - \frac{X_{2k-1} + X_{2k}}{2} \right)^2 + \left(X_{2k} - \frac{X_{2k-1} + X_{2k}}{2} \right)^2.$$

Note that each Y_k is an example of V_2' , an estimate of the variance of X using two samples, given in Theorem 10.13. Show that if $E[X^k] < \infty$ for $k = 1, 2, 3, 4$, then the sample mean $M_n(Y)$ is a consistent, unbiased estimate of $\text{Var}[X]$.

10.4.6 ♦♦ An experiment produces a Gaussian random vector $\mathbf{X} = [X_1 \ \cdots \ X_k]'$ with $E[\mathbf{X}] = \mathbf{0}$ and correlation matrix $\mathbf{R} = E[\mathbf{X}\mathbf{X}']$. To estimate $\mathbf{R}$, we perform n independent trials, yielding the iid sample vectors $\mathbf{X}(1), \mathbf{X}(2), \dots, \mathbf{X}(n)$, and form the sample correlation matrix

$$\hat{\mathbf{R}}(n) = \frac{1}{n} \sum_{m=1}^n \mathbf{X}(m) \mathbf{X}'(m).$$

- (a) Show $\hat{\mathbf{R}}(n)$ is unbiased by showing $E[\hat{\mathbf{R}}(n)] = \mathbf{R}$.
- (b) Show that the sequence of estimates $\hat{\mathbf{R}}(n)$ is consistent by showing that every element $\hat{R}_{ij}(n)$ of the matrix $\hat{\mathbf{R}}$ converges to R_{ij} . That is, show that for any $c > 0$,

$$\lim_{n \rightarrow \infty} P \left[\max_{i,j} |\hat{R}_{ij} - R_{ij}| \geq c \right] = 0.$$

Hint: Extend the technique used in Problem 10.4.4. You will need to use the result of Problem 7.6.4 to show that $\text{Var}[X_i X_j]$ is finite.

10.5.1 ● $X_1, \dots, X_n$ are n independent identically distributed samples of random

variable X with PMF

$$P_X(x) = \begin{cases} 0.1 & x = 0, \\ 0.9 & x = 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) How is $E[X]$ related to $P_X(1)$?
 (b) Use Chebyshev's inequality to find the confidence level α such that $M_{90}(X)$, the estimate based on 90 observations, is within 0.05 of $P_X(1)$. In other words, find α such that

$$P[|M_{90}(X) - P_X(1)| \geq 0.05] \leq \alpha.$$

- (c) Use Chebyshev's inequality to find out how many samples n are necessary to have $M_n(X)$ within 0.03 of $P_X(1)$ with confidence level 0.1. In other words, find n such that

$$P[|M_n(X) - P_X(1)| \geq 0.03] \leq 0.1.$$

10.5.2 ● X is a Bernoulli random variable with unknown success probability p . Using 100 independent samples of X , find a confidence interval estimate of p with confidence coefficient 0.99. If $M_{100}(X) = 0.06$, what is our interval estimate?

10.5.3 ● In n independent experimental trials, the relative frequency of event A is $\hat{P}_n(A)$. How large should n be to ensure that the confidence interval estimate

$$\hat{P}_n(A) - 0.05 \leq P[A] \leq \hat{P}_n(A) + 0.05$$

has confidence coefficient 0.9?

10.5.4 ■ When we perform an experiment, event A occurs with probability $P[A] = 0.01$. In this problem, we estimate $P[A]$ using $\hat{P}_n(A)$, the relative frequency of A over n independent trials.

- (a) How many trials n are needed so that the interval estimate

$$\hat{P}_n(A) - 0.001 < P[A] < \hat{P}_n(A) + 0.001$$

has confidence coefficient $1 - \alpha = 0.99$?

- (b) How many trials n are needed so that the probability $\hat{P}_n(A)$ differs from $P[A]$ by more than 0.1% is less than 0.01?

10.6.1 ● Graph one trace of the sample mean of the Poisson (1) random variable. Calculate (using a central limit theorem approximation) and graph the corresponding 0.9 confidence interval estimate.

10.6.2 ● X is the Bernoulli (1/2) random variable. The sample mean $M_n(X)$ has standard error

$$e_n = \sqrt{\frac{\text{Var}[X]}{n}} = \frac{1}{2\sqrt{n}}.$$

The probability that $M_n(X)$ is within one standard error of p is

$$p_n = P\left[\frac{1}{2} - \frac{1}{2\sqrt{n}} \leq M_n(X) \leq \frac{1}{2} + \frac{1}{2\sqrt{n}}\right].$$

Use the `binomialcdf` function to calculate the exact probability p_n as a function of n . What is the source of the unusual sawtooth pattern? Compare your results to the solution of Quiz 10.6.

10.6.3 ■ Recall that an exponential (λ) random variable X has

$$E[X] = 1/\lambda, \\ \text{Var}[X] = 1/\lambda^2.$$

Thus, to estimate λ from n independent samples $X_1, \dots, X_n$, either of the following techniques should work.

- (a) Calculate the sample mean $M_n(X)$ and form the estimate $\hat{\lambda} = 1/M_n(X)$.
 (b) Calculate the unbiased variance estimate $V'_n(X)$ of Theorem 10.13 and form the estimate $\tilde{\lambda} = 1/\sqrt{V'_n(X)}$.

Use MATLAB to simulate the calculation $\hat{\lambda}$ and $\tilde{\lambda}$ for $m = 1000$ experimental trials to determine which estimate is better.

10.6.4 ♦ $\mathbf{X}$ is 10-dimensional Gaussian ($\mathbf{0}, \mathbf{I}$) random vector. Since $E[\mathbf{X}] = \mathbf{0}$, $\mathbf{R}_{\mathbf{X}} = \mathbf{C}_{\mathbf{X}} = \mathbf{I}$. We will use the method of Problem 10.4.6 and estimate $\mathbf{R}_{\mathbf{X}}$ using the

sample correlation matrix

$$\hat{\mathbf{R}}(n) = \frac{1}{n} \sum_{m=1}^n \mathbf{X}(m) \mathbf{X}'(m).$$

For $n \in \{10, 100, 1000, 10,000\}$, construct a MATLAB simulation to estimate

$$\mathrm{P} \left[\max_{i,j} \left| \hat{R}_{ij} - I_{ij} \right| \geq 0.05 \right].$$

10.6.5 ♦ In terms of parameter a , random variable X has CDF

$$F_X(x) = \begin{cases} 0 & x < a - 1, \\ 1 - \frac{1}{[x - (a - 2)]^2} & x \geq a - 1. \end{cases}$$

- (a) Show that $\mathrm{E}[X] = a$ by showing that $\mathrm{E}[X - (a - 2)] = 2$.
 - (b) Generate $m = 100$ traces of the sample mean $M_n(X)$ of length $n = 1000$. Do you observe convergence of the sample mean to $\mathrm{E}[X] = a$?
-